

| Method                          | Short        | Medium       | Long         | Entire dataset |
|---------------------------------|--------------|--------------|--------------|----------------|
| First                           | 46.0%        | 40.1%        | 39.4%        | 41.9%          |
| Center                          | 47.9%        | 40.0%        | 42.6%        | 43.5%          |
| FPS:2:4:600                     | <b>72.6%</b> | 58.7%        | 49.9%        | 60.4%          |
| MaxInfo:1000:600, R=8, tol=0.3  | 68.1%        | 54.9%        | 49.6%        | 57.5%          |
| MaxInfo:2000:600, R=8, tol=0.3  | 69.1%        | 55.6%        | 49.6%        | 58.1%          |
| MaxInfo:1000:600, R=8, tol=0.15 | 70.3%        | 59.6%        | <b>51.1%</b> | 60.3%          |
| MaxInfo:1000:600, R=12, tol=0.3 | 70.2%        | 59.1%        | 49.7%        | 59.7%          |
| CSTA:1000:600, without KTS      | 70.2%        | 57.7%        | 50.3%        | 59.4%          |
| CSTA:2000:600, without KTS      | 71.1%        | <b>61.0%</b> | 49.3%        | <b>60.5%</b>   |
| CSTA:1000:600, with KTS         | 60.3%        | 53.9%        | 48.1%        | 54.4%          |

Table 9: VideoMME results over frame sampling techniques: Model: Qwen2.5

| Method                         | Short        | Medium       | Long         | Entire dataset |
|--------------------------------|--------------|--------------|--------------|----------------|
| First                          | 46.0%        | 40.1%        | 39.4%        | 41.9%          |
| Center                         | 47.9%        | 40.0%        | 42.6%        | 43.5%          |
| FPS:2:4:96                     | <b>71.9%</b> | <b>61.3%</b> | 49.4%        | <b>60.9%</b>   |
| MaxInfo:1000:96, R=8, tol=0.3  | 68.1%        | 54.9%        | 49.6%        | 57.5%          |
| MaxInfo:2000:96, R=8, tol=0.3  | 69.1%        | 55.6%        | 49.6%        | 58.1%          |
| MaxInfo:1000:96, R=8, tol=0.15 | 68.2%        | 56.0%        | <b>50.2%</b> | 58.1%          |
| MaxInfo:1000:96, R=12, tol=0.3 | 69.8%        | 59.3%        | 49.6%        | 59.6%          |
| CSTA:1000:96, without KTS      | 68.9%        | 55.7%        | 49.1%        | 57.9%          |
| CSTA:2000:96, without KTS      | 68.6%        | 57.0%        | 49.1%        | 58.2%          |
| CSTA:1000:96, with KTS         | 59.5%        | 51.6%        | 46.6%        | 52.9%          |

Table 10: VideoMME results over frame sampling techniques: Model: Qwen2.5